TARUN VUSIRIKAYALA

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PROFESSIONAL SUMMARY

Senior Data Analyst *with experience in healthcare and financial services, specializing in SQL, Python, ETL pipelines, Power BI and predictive modeling. Demonstrated expertise in claims analysis, credit risk assessment and data quality optimization. Proven track record of transforming complex datasets into actionable insights, delivering measurable business impact including cost savings, operational improvements, and enhanced decision-making across claims processing, member analytics, and risk management.*

TECHNICAL SKILLS

Programming & Analytics: SQL, Python, R, SAS, Java, C, C#, C++, Predictive Modeling, Statistical Analysis, A/B Testing
Databases & Warehousing: Oracle, Azure SQL Database, Azure Synapse Analytics, Snowflake, Microsoft SQL Server, PostgreSQL, AWS Redshift, Google BigQuery
Data Integration & ETL: Azure Data Factory, Talend, Alteryx, ETL Pipelines, Data Cleaning, Data Validation, Reconciliation
Business Intelligence & Visualization: Power BI, Tableau, QlikView, KPI Dashboards, Reporting Automation
Cloud Tools & Platforms: Microsoft Azure, AWS, GCP, Git, GitHub, Docker
Analytics Automation & Productivity: Advanced Excel (Power Query, Pivot Tables, VLOOKUP), VBA (Excel Automation), Microsoft 365
AI & ML: Pandas, NumPy, Scikit-learn, Statsmodels, TensorFlow, PyTorch, Azure OpenAI, LangChain, RAG, LLMs, Agents
Project & Delivery Methodologies: Agile/Scrum, SDLC, Jira, Waterfall
Healthcare Analytics & Compliance: Claims Data, Patient Analytics, Drug/Pharmacy Analysis, Adherence Metrics, Risk Adjustment
Financial Domain: Credit Risk, AML/KYC, Fraud Analytics, Regulatory Reporting, Customer Segmentation, Default Prediction, Transaction Analysis

PROFESSIONAL EXPERIENCE

Senior Data Analyst, Cigna Healthcare | March 2025 – Present

• Analyzed medication adherence patterns across 180K+ members utilizing SQL and Python to drive HEDIS compliance and quality metrics; segmented high-risk non-adherent populations by clinical diagnosis and deployed cohorts leveraging Azure cloud infrastructure, achieving 25% improvement in diabetes and hypertension adherence rates and supporting Star Ratings performance.

• Evaluated claims processing inefficiencies through SQL and Tableau dashboards; identified 60% of manual denials originated from incomplete prior authorization documentation; redesigned intake workflow with automated status notifications in Agile/Scrum environment using Azure data pipelines, reducing average resolution time from 8 to 3 days and decreasing escalations by 50%.

• Performed comprehensive pharmacy cost analysis on specialty drug utilization patterns using Python statistical modeling and Statsmodels for advanced hypothesis testing; identified 15% of spend attributed to off-label prescriptions without clinical justification; partnered with medical director and pharmacy benefit managers on prior authorization guideline updates, generating net savings while maintaining access to medically necessary medications.

• Developed and deployed predictive member risk segmentation model leveraging claims history, comorbidity burden, and demographic variables using Python with advanced machine learning techniques; operationalized segmentation across sales and retention teams using cloud-based infrastructure, improving plan renewal rates by 3.2% and supporting targeted member retention strategies.

• Designed and implemented automated operational dashboards in Tableau tracking real-time claims volume, enrollment transactions, and pharmacy metrics stratified by processor; collaborated with SDLC teams in Jira-managed Agile sprints to eliminate ad-hoc reporting requests weekly and enabled same-day anomaly detection and process monitoring.

Data Analyst, HSBC Global Banking & Markets | January 2021 – November 2022

▪ Developed ETL workflows using Talend and SQL to consolidate customer master data from fragmented legacy systems into Snowflake data warehouse; standardized data definitions across 15+ regional offices reducing duplicate records by 45% and improving downstream reporting accuracy for relationship managers.

▪ Created operational dashboards in Tableau and Power BI tracking lending pipeline velocity, approval turnaround times, and underwriting bottlenecks; provided weekly insights to Credit Management team enabling faster decision cycles on corporate lending deals.

▪ Conducted statistical analysis using Python to model customer credit profiles and predict default risk; findings informed updates to credit policy thresholds and reduced unexpected charge-offs in mid-market segment by identifying early warning indicators.

▪ Automated customer data validation and onboarding workflows using Python and SQL; established data quality controls that decreased manual corrections by 30% and accelerated time-to-revenue for new account activations.

▪ Queried Microsoft SQL Server databases to extract payment transaction volumes and settlement patterns; analyzed cross-border flow data supporting Treasury team's working capital management and cash positioning strategies across multiple currencies.

▪ Partnered with operations and finance teams to identify process inefficiencies in reconciliation procedures; delivered analytical recommendations that streamlined reporting infrastructure and strengthened internal control documentation for regulatory audits.

Data Science Intern, Modak |June 2020 - December 2020

▪ Developed predictive models using Python (Pandas, NumPy, Scikit-learn) to analyze user spending patterns and segment teen customers by financial behavior; insights informed product roadmap decisions and supported 18% improvement in card onboarding conversion rates.

▪ Performed exploratory data analysis on transactional and app engagement data using SQL; identified churn drivers and onboarding friction points enabling product team to reduce day-7 user drop-off by 12%.

▪ Designed and executed A/B testing framework for gamification features; conducted statistical analysis on engagement metrics to validate hypothesis on reward mechanics effectiveness across different user demographics.

PROJECTS

Sentiment Analysis of Drug Reviews (Python, NLP, Scikit-learn, VADER, Naive Bayes, NLTK, Pandas, TF-IDF, Matplotlib)

▪ Implemented sentiment analysis on drug reviews to classify patient conditions using NLP techniques, enhancing healthcare decision-making using Naive Bayes and VADER algorithms, demonstrating proficiency in Python and machine learning libraries.

▪ Expertly managed data preprocessing and feature engineering, including text data cleaning, bag-of-words, and TF-IDF vectorization, to optimize machine learning models for healthcare insights.

Automating Data Visualizations with Generative AI: (Python, SQL, Streamlit, OpenAI GPT-4, Plotly, Seaborn, Matplotlib)

▪ Built an AI-powered self-service analytics tool using Python, SQL, and GPT-4 to auto-generate EDA insights, trends, distributions, and outliers from CSV datasets

▪ Automated EDA and reporting workflows using Python with Plotly and Matplotlib, reducing manual analysis effort and ad-hoc reporting time.

EDUCATION

SUNY Institute of Technology *| Master of Science, Computer Science | Jan 2023 - Dec 2024 | Utica, New York*

CVR College of Engineering *| Bachelor of Technology, Computer Science | Aug 2018 - Apr 2022 | Hyderabad, India*